

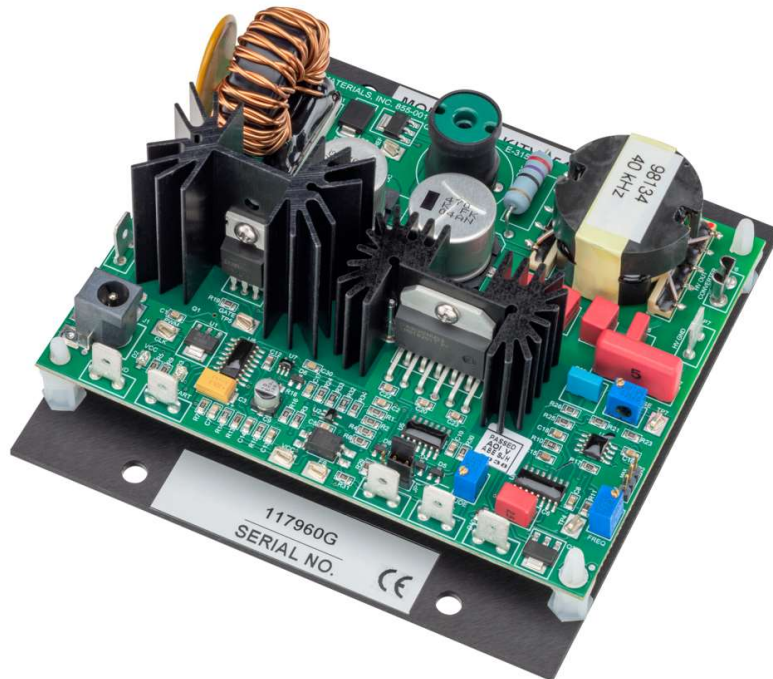


SONICS.com[®]
SONICS & MATERIALS, INC.

ULTRASONIC PROCESSOR

Part No. KITVC544

MANUAL



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1. WARRANTY

Your Ultrasonic Processor is warranted and backed by the manufacturer for a period of 6 months from the date of shipment against defects in material and workmanship under normal use as described in this instruction manual. During the warranty period, the manufacturer will, at its option, as the exclusive remedy, either repair or replace without charge for material and labor, the part(s) which prove to be defective, provided the unit is returned to us properly packed with all transportation charges prepaid.

The manufacturer neither assumes nor authorizes any person to assume for it any other obligations or liability in connection with the sale of its products. No person or company is authorized to change, modify, or amend the terms of this warranty in any manner or fashion whatsoever. Under no circumstances shall the manufacturer be liable to the purchaser or any other person for any incidental or consequential damages or loss of good will, production, or profit resulting from any malfunction or failure of its product.

This warranty does not apply to equipment that has been subject to unauthorized repair, misuse, abuse, negligence or accident. Equipment which shows evidence of having been used in violation of operating instructions, or which has had the serial number altered or removed, will be ineligible for service under this warranty.

The aforementioned provisions do not extend the original warranty period of any product that has either been repaired or replaced by the manufacturer.

2. WARNINGS

Please read the manual in its entirety. Necessary instruction and guidance are provided to help ensure the successful operation of this device. Observe the following:

- High voltage is present in the power supply, converter and high frequency cable. There are no user-serviceable parts inside any of these devices. Do NOT attempt to remove the converter case.
- Power supply negative connection (+12 VDC /+24 VDC GND) shall be connected to earth ground.
- Do NOT touch any open cable connections on the unit while the power is turned ON.
- Do NOT operate power supply with converter disconnected from high voltage cable. High voltage is present in the cable and may pose a shock hazard.
- Do NOT attempt to disconnect the converter high voltage cable while the unit is running.
- Install the ultrasonic power supply in an area free from excessive dust, dirt, explosive or corrosive fumes and protected from extremes in temperature and humidity. (See page 5 for specifications) Do not place the power supply within a Fume Hood.
- NEVER immerse the entire converter in liquids of any kind, or let condensed moisture or liquid drip into the converter.
- NEVER grasp an activated horn or touch the tip of a vibrating probe. It can cause severe burns and tissue damage.
- NEVER hold or clamp the converter by the front driver or by the probe itself. This can cause permanent damage to the system. Support the converter by only clamping around the converter housing (upper portion). If in doubt about the area to be clamped, contact the manufacturer.
- Inspect high frequency cable for cracks in the protective outer jacket.
- Do not operate unit with a damaged cable. Doing so may cause serious injury.

Symbols



Caution, Risk of electric shock, Hazardous voltage.



Caution, Risk of danger. Refer to User Manual.

3. SPECIFICATIONS

Power Supply	KITVC544
Input Voltage	12 VDC $\pm$ 0.5 VDC @ 5 AMPS or 24VDC $\pm$ 0.5 VDC @ 2.5 AMPS
Electrical Output	High Voltage out: P6 (Red converter wire) High Voltage Ground: P7 (Blue Converter Wire)
Weight (total)	0.702 pounds (318 grams) $\pm$ 10%
Dimensions (PCB)	4.25"W x4.75"L x 2.035"H (108.0mm x 120.7mm x 51.7mm)
Mounting	See Section 6. Mounting for mounting hole locations and size
Output Voltage	380 VRMS (max.)
Output Frequency	40 KHz

Environmental	
Pollution Degree	2
Installation Category	II
Operating Limits	Temperature: 32 - 104°F (0°C to 40°C) Relative Humidity 10 - 95% (Non Condensing) Altitude: 6,651 ft. (2000 m)
Shipping/Storage	Temperature: 35 -120 °F (2 - 49 °C) Relative Humidity 10 - 95% (Non Condensing) Ambient Pressure Extremes: 40,000 ft. (12,192 m)
Restriction of Hazardous Substances (ROHS)	RoHS Compliant Directive 2002/95/EC
Relative humidity	Maximum relative humidity 80% for temperatures up to 31°C decreasing linearly to 50% relative humidity to 40°C
Other	For indoor use only



WEEE Statement

This product contains electrical or electronic materials. The presence of these materials may, if not disposed of properly, have potential adverse effects on the environment and human health. Presence of this label on the product means it should not be disposed of as unsorted waste and must be collected separately. As a consumer, you are responsible for ensuring that this product is disposed of properly. To find out how to properly dispose of this product contact Customer Service.

4. PRINCIPLES OF OPERATION

The ultrasonic power supply transforms the input power to a specific 40 KHz signal that drives a piezoelectric converter/transducer. This electrical signal is converted by the transducer to a mechanical vibration due to the characteristics of the internal piezoelectric crystals.

The vibration is amplified and transmitted down the length of the horn/probe where the tip longitudinally expands and contracts. The distance the tip travels is dependent on the amplitude set on the power supply. As you increase the amplitude setting the sonication intensity will increase.

In liquid, the rapid vibration of the tip causes cavitation, the formation and violent collapse of microscopic bubbles. The collapse of thousands of cavitation bubbles releases tremendous energy in the cavitation field. The erosion and shock effect of the collapse of the cavitation bubble is the primary mechanism of fluid processing.

Relationship of Amplitude and Wattage

Ultrasonic power is measured in watts. Amplitude is a measurement of the excursion (displacement peak to peak) of the tip of the probe or the front driver (a probe is also known as a horn).

The ultrasonic power supply is designed to deliver constant amplitude regardless of changes in load. As a liquid is processed, the load on the probe will vary due to changes in the liquid sample (i.e. viscosity, concentration, temperature, etc.). As the resistance to the movement of the probe increases (increased load on the probe), additional power will be delivered by the power supply to ensure that the excursion at the probe tip remains constant. The wattage readings will vary as the load changes; however, the amplitude will remain the same.

The resistance to the movement of the probe determines how much power will be delivered to maintain amplitude.

The amplitude control allows the ultrasonic vibrations at the probe tip to be set to any desired level but the amount of power required cannot be predetermined. A sensing network continuously monitors the output requirements, and automatically adjusts the power to maintain the amplitude at the preselected level. The greater the resistance to the movement of the probe due to higher viscosity, deeper immersion of the probe into the sample, larger probe diameter or higher pressure, the greater the amount of power that will be delivered to the probe. Setting the amplitude control to its maximum will not cause the maximum power rating of the unit to be delivered to the sample. The maximum power that the ultrasonic power supply is capable of delivering will only be achieved when the resistance to the movement of the probe is high enough to draw maximum wattage.

It is the intensity of cavitation that measures the effectiveness of the sonication, not the total power applied to the system. Intensity is directly related to the amplitude of the radiating face of the tip or horn. It is amplitude that must be provided, maintained, and monitored. The unit provides controlled amplitude under varying load conditions in order to give reproducible results.

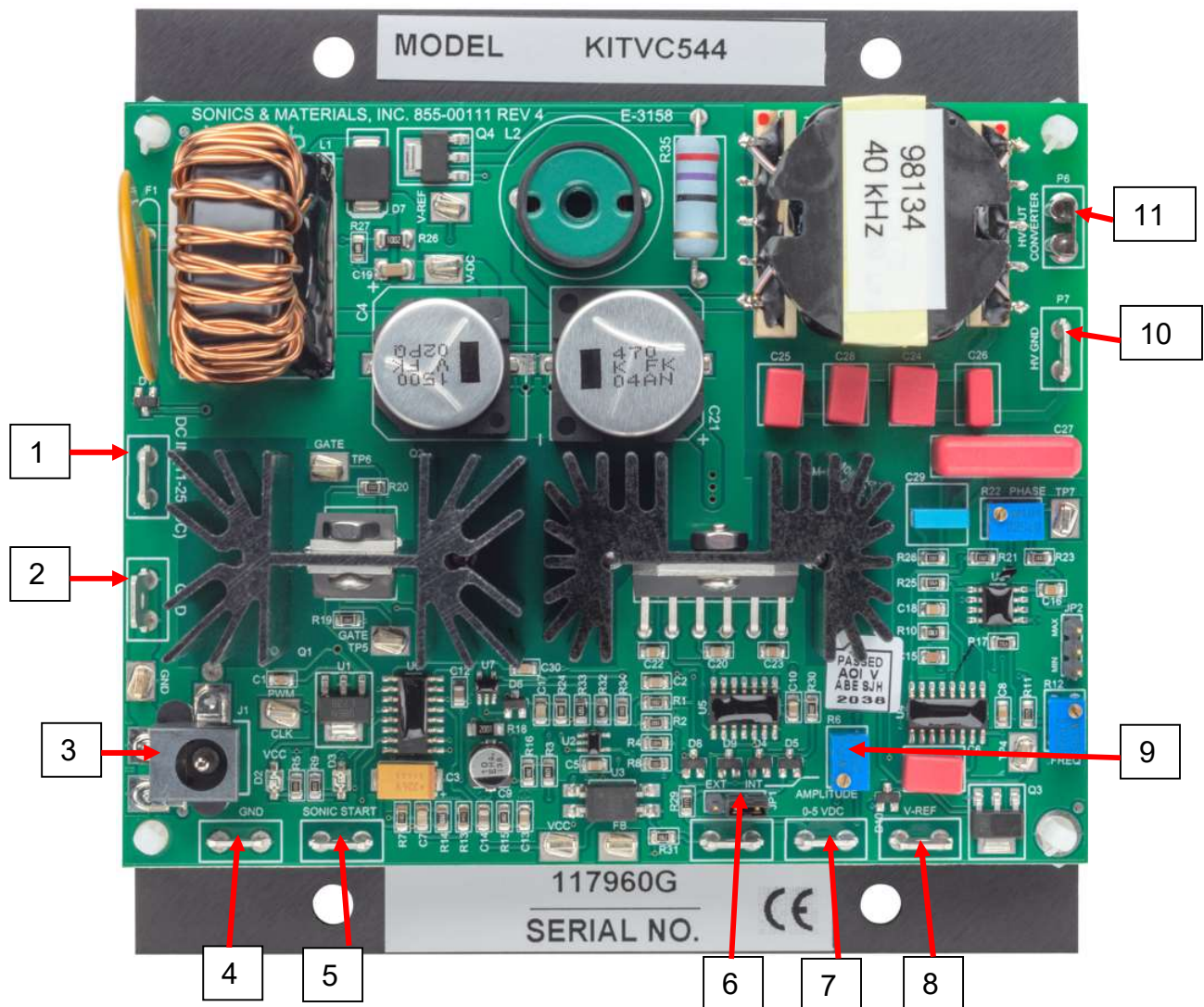
5. OPERATION & CONTROLS

5.1. Electrical Connections & Controls

See the diagram on the next page for the physical connection locations.

Location	Name	Function
1	DC Power Input	Input Power Connection 12.0 $\pm$ 0.5 VDC / 24.0 $\pm$ 0.5 VDC, Power = 60 Watts, Internal Fuse Protected at 7A, Auto Resettable
2	Ground Power Input	Input Control Connection GND = ON, Open = OFF ON Resistance less than 0.3 Ohms OFF Resistance greater than 10 Mega Ohm
3	Power Plug	Alternate power input connection (replaces 1 & 2) This should only be used with a 24 VDC Input Voltage
4	Sonic Ground Input	Ground connection for use with the "Sonic Start" input.
5	Sonic Start Input	To activate ultrasonics, first place the board on 'stand-by' by applying DC Power, then connect terminal "Sonic Start" to terminal "Sonic Ground Input". Ultrasonics will stay on as long as the terminals are connected together. Do not activate and deactivate ultrasonics by turning the input power on and off, as it will cause the board to fail.
6	(JP1)	Internal or external amplitude control connector with jumper INT = Amplitude controlled by adjusting potentiometer R6. Ext = Amplitude controlled by using an external Potentiometer.
7	External Amplitude Control Voltage	Input Control Signal – Signal Controls the Amplitude (displacement) of the converter. 2 to 5 VDC, 2 VDC = 40%, 5 VDC = 100%, Linear Scale, Input Impedance > 5K Ohms Note: A 10 K ohm potentiometer connected from V-Ref (Location 8) to Gnd (Location 2 or 4) with the wiper connected to Amplitude (Location 7) can be used to adjust the amplitude externally. Right under the #6 arrow is a GND for the low side of the External Amplitude POT. (Only active when the left and center pins of JP1 are connected with the jumper) Example: For 12 VDC Operation, using a 15.0 μ m converter, an Amplitude Control Voltage of 2 VDC will produce approximately 15 μ m * 0.4 = 6.0 μ m displacement (40% being the minimum possible amplitude setting).
8	Reference Voltage	Reference voltage for external amplitude control. 5.1 VDC reference.
9	Int. Amplitude Ctrl (on board)	Integrated onboard amplitude control potentiometer, R6 (only active when the center and right pins of JP1 are connected with the jumper) Use V-REF as positive and GND as negative for monitoring of Amplitude (displacement) of the converter. 2 to 5 VDC, 2 VDC = 40%, 5 VDC = 100%, Linear Scale ***Do NOT adjust any other PCB controls***
10	High Voltage Ground (P7)	This is the ground connection for the converter. Connect to the blue lug on the converter cable (the shield).

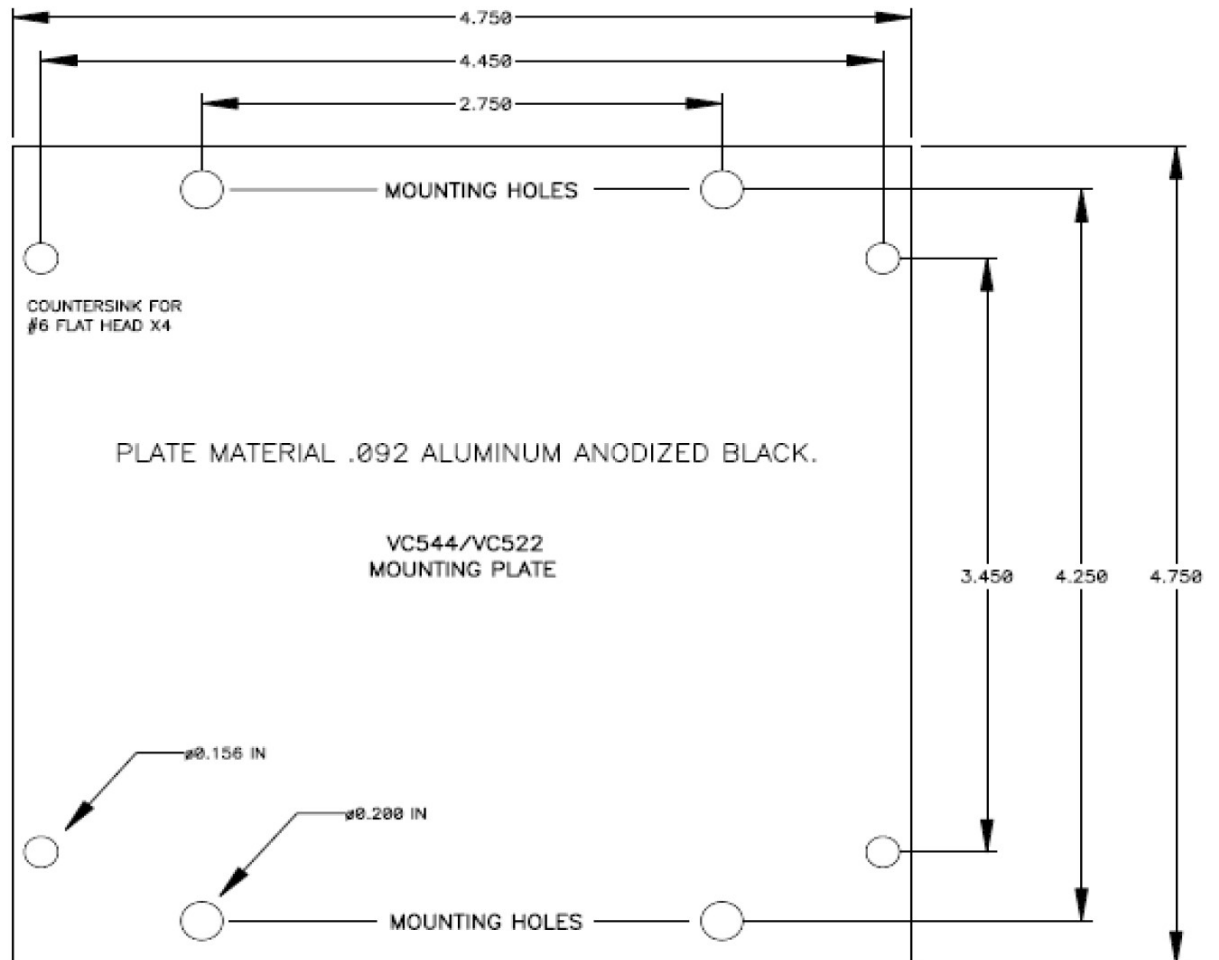
11	High Voltage Out (P6)	This is the high voltage output to the converter. Connect to the red lug on the converter cable (the center conductor).
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6. MOUNTING

Note: Converter cables should have a minimum bend radius of 10 to 15 times the cable diameter.

All dimensions in inches. Tolerance: ± 0.010 inches



7. APPLICATIONS

It is recommended to energize the ultrasonics for a minimum of 1 second, and maximum of approximately 30 seconds continuously. Longer processing times are possible depending on the amount of force used against the converter/probe unit. Contact us to discuss longer potential processing times.

Each of these variables should be discussed with your Sonics representative in order to reduce optimization time and achieve good results:

1. **Amplitude** – the peak to trough distance which the converter face travels, in the axial direction, when ultrasonics is energized. Amplitude is adjustable within a given range determined by the converter. Higher amplitude may be needed to obtain desired results, but will also create more heat.
2. **Contact Force** – the force applied by the converter to the sample vessel. Sufficient force is needed for proper coupling of the converter to the vessel.
3. **Contact Area** – maximize the surface between the sample vessel and the converter. For example, a round surface against a flat vessel is not recommended.
4. **Duration** – time that ultrasonics is energized and processing the sample. Trials must be done to determine the optimal duration.
5. **Sample Vessel** – different materials will have a great effect on how well the ultrasonic energy is transferred through, and into the liquid sample. Wall thickness and shape should be discussed and different options tested.
6. **Sample Volume** – smaller volumes can generally be processed more quickly.

To save time, please consult with a Sonics representative and discuss all aspects of the application and sample vessel.



8. RETURN OF EQUIPMENT

Ultrasonic equipment in need of repair should be sent back to the factory.

In order to receive prompt service, contact your Customer Service Representative before returning any instrument.

You must obtain a Return Authorization Number (RMA) prior to returning the equipment.

Care should be exercised to provide adequate packing to insure against possible damage in shipment. The Ultrasonic Processor should be sent to the address below with all transportation charges prepaid and return of shipment indicated.

RMA # -----
Sonics & Materials
53 Church Hill Rd.
Newtown, CT 06470

Important

The user must certify that the ultrasonic processor and/or the accessories returned for repair are free of any biohazardous or radioactive material and are safe for handling. Please complete the "Safety Certification" form on the next page and send it in with your equipment.

Do not return any equipment unless such a certification can be made.

9. SAFETY CERTIFICATION FORM

Items being returned:

Please check only one item below:

☐ The equipment was never used or exposed to any radiological, biological or chemical agents and is safe to handle, use or dispose of.

☐ The equipment was used but not in conjunction with or exposed to any radiological, geological or chemical agents and is safe to handle, use, or dispose of.

☐ The equipment was used in conjunction with or exposed to radiological, biological, or chemical agents and has been decontaminated, rendering it safer for handling, use, or disposal.

Authorization

By accepting authorization to return the equipment listed above, the undersigned assumes all responsibility and liability for radiological, biological and chemical decontamination. Delivery of the equipment can be refused if necessary documentation is not provided or where it is determined that the equipment has not been properly decontaminated. If it is determined that the equipment was not properly decontaminated, the Authorized Repair Facility reserves the right to bill the customer for any and all costs associated with the decontamination and/or appropriate disposal of the equipment. In the event the equipment has been exposed to radiological contamination, the signature of the Radioactive Safety Officer is required.

Print name: _____ RMA # _____

Signature: _____ Date: _____